DATA 602: Statistical Data Analysis

Analyzing Student Participation Patterns and Academic Performance Across Different Levels of Absenteeism

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1. Purpose

1.1 Domain

The domain this project focuses on is education and data mining, specifically the use of student attendance, engagement, and academic-performance data to understand patterns in e-learning environments. The project examines whether routinely recorded learner activity can provide useful information about differences in student academic performance.

1.2 Investigation Goal

The goal of this investigation was to determine whether student absenteeism (categorized as fewer than 7 absences vs. more than 7 absences) is associated with student engagement and academic performance among students?

a) Guiding question 1: Is the proportion of high-level performers in the low-absence group greater than the proportion of high-level performers in the high-absence group
b) Guiding question 2: Is the average number of times course content is visited greater among students with low absenteeism than among students with high absenteeism?
c) Guiding question 3: Is the average number of times discussions are visited higher among students with low absenteeism than among students with high absenteeism?
d) Guiding question 4: Is the standard deviation of the number of hands raised in class different among students with low absenteeism than among students with high absenteeism?

1.3 Practical Implications

Absenteeism is one of the easiest metrics for a teacher to record in any school setting. If absenteeism is shown to be meaningfully associated with lower performance/engagement, schools can use attendance records alongside other metrics to assist at-risk students before they begin to perform poorly.

1.4 Population(s) of Interest:

The target population is students in grades G-01 through G-12 that are enrolled in courses that are delivered through the Kalboard 360 e-learning program. The population spans multiple nationalities, genders, and course areas.

2. Data

2.1 Source of Data

Name: Students' Academic Performance Dataset (xAPI-Edu-Data)

Website: [Students' Academic Performance Dataset]

Original Data Source/Authors: Elaf Abu Amrieh, Thair Hamtini, and Ibrahim Aljarah, The University of Jordan, Amman, Jordan.

Data Collection Platform: Kalboard 360 learning management system (LMS)

2.2 Permission/Accessibility

Under CC BY-SA 4.0 license, allows sharing in any medium or format (commercially is allowed). Allows any remix, transformation, or build on material (commercially is allowed) as long as proper attribution is made, including providing a link to the license and distributing the material under the same license.

2.3 Time Frame of Data Collection

The dataset consists of 480 student records collected across two semesters (a First and Second semester) of a single school year at the time of the original study's publication (2015–2016), spanning multiple grade levels and course subjects taught through Kalboard 360. No end or start times of the semesters were given.

2.4 Methodology

The data is collected using a learner activity tracker tool, which is called experience API (xAPI). The xAPI is a component of the training and learning architecture (TLA) that enables monitoring learning progress and learner's actions like reading an article or watching a training video. The experience API helps the learning activity providers to determine the learner, activity and objects that describe a learning experience.

2.5 Variables of Interest

Variable name	Description	Type
StudentAbsenceDays	The number of absence days for each student	Categorical (Above-7, Under-7)
Class	The grade a student achieves in a course, classified into 3 intervals	Categorical. Low-Level: 0 to 69; Middle-Level: 70 to 89; High-Level: 90 to 100
VisITedResources	How many times the student visits course content	Numerical, continuous (0 to 100)

raisedHands	How many times the student raises their hand in the classroom	Numerical, continuous (0 to 100)
Discussion	How many times the student participates in discussions groups	Numerical, continuous (0 to 100)

3. Topics

3.1 (a) Is the proportion of high-level performers in the low-absence group greater than the proportion of high-level performers in the high-absence group?

Let p_{HN} = the proportion of high-level performers among students with fewer than 7 absent days.

Let p_{HY} = the proportion of high-level performers among students with more than 7 absent days.

$$H_0: p_{HN} \leq p_{HY}$$

$$H_A: p_{HN} > p_{HY}$$

Rejecting H_0 indicates that a greater proportion of students with fewer than 7 absence days reach the High-performance band than students with more than 7 absence days. Failing to reject H_0 would mean these data provide no evidence of such a difference.

3.2 b) Is the average number of times course content is visited greater among students with low absenteeism than among students with high absenteeism?

Let μ_{LC} = mean number of times course content is visited by students with low absenteeism

Let μ_{HC} = mean number of times course content is visited by students with high absenteeism

$$H_0: \mu_{LC} \leq \mu_{HC}$$

$$H_A: \mu_{LC} > \mu_{HC}$$

Rejecting H_0 would indicate that students with fewer than 7 absence days visit course content more often on average than students with more than 7 absence days. Failing to reject would mean these data provide no evidence of such a difference.

3.3 c) Is the average number of times discussion posts are visited higher among students with low absenteeism than among students with high absenteeism?

Let μ_{LD} = mean number of times discussion posts are visited by students with low absenteeism

Let μ_{HD} = mean number of times discussion posts are visited by students with high absenteeism

$$H_0: \mu_{LD} \leq \mu_{HD}$$

$$H_A: \mu_{LD} > \mu_{HD}$$

Rejecting H_0 would indicate that students with fewer than 7 absence days visit discussion posts more often on average than students with more than 7 absence days. Failing to reject H_0 would mean these data provide no evidence that students with fewer than 7 absence days visit discussion posts more often on average.

3.4 d) Is the standard deviation of the number of hands raised in class different among students with low absenteeism than among students with high absenteeism?

Let σ_{LH} = standard deviation of the number of hands raised among students with low absenteeism

Let σ_{HH} = standard deviation of the number of hands raised among students with high absenteeism

$$H_0: \sigma_{LH} = \sigma_{HH}$$

$$H_A: \sigma_{LH} \neq \sigma_{HH}$$

Rejecting H_0 would indicate that the variability in the number of hands raised is different between students with fewer than 7 absence days and students with more than 7 absence days. Failing to reject H_0 would mean these data provide no evidence of a difference in the standard deviations.

4. Statistical Methods

4.1 Guiding Question 1

The hypotheses for this question are given in Section 3.1(a). A permutation test was used with 1,999 random permutations at a significance level of $\alpha = 0.05$.

Among students with low absenteeism, 138 of 289 (47.75%) were High-level performers, compared with 4 of 191 (2.09%) among students with high absenteeism. The observed difference in proportions was (See appendix 6.1.1)

$$\hat{p}_{HN} - \hat{p}_{HY} = 0.4566$$

The permutation test gave an empirical P-value of 0.0005. Since $P < 0.05$, the null hypothesis was rejected. As shown in Figure 1, none of the permuted differences approached the observed difference, indicating that the observed result was unlikely under the null hypothesis. (See appendix 6.1.2)

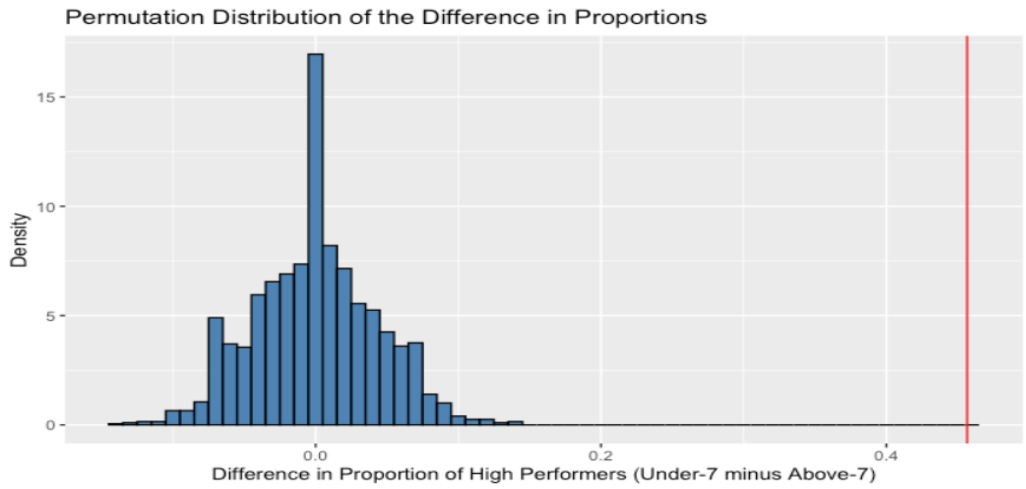


Figure 1. Permutation Distribution of the Difference in Proportions (See appendix 6.1.4)

The 95% confidence interval for the difference in proportions, computed using the plus-four adjustment, was (See appendix 6.1.3)

$$0.3901 < p_{HN} - p_{HY} < 0.5134$$

From these data, we are 95% confident that the difference in the proportion of High-level performers between the low- and high-absence groups is between 39.01 and 51.34 percentage points.

Overall, the results provide strong evidence that students with fewer than 7 absent days had a higher proportion of High-level performers than students with more than 7 absent days. However, because the data are observational, this association does not imply that absenteeism causes lower academic performance.

4.2 Guiding Question 2

The hypotheses are stated in Section 3.2(b). A Welch two-sample t-test was conducted at α = 0.05.

Students with low absenteeism (<7 days, n = 289) averaged 68.20 course content visits (SD = 27.61), compared with 34.51 visits (SD = 30.27) for students with high absenteeism (n = 191). The observed mean difference was 33.69 visits (Appendix 6.2.1).

The Welch t-test showed a significant difference, t(380.51) = 12.356, p < 2.2e-16, so the null hypothesis was rejected. Figure 2 also shows higher engagement among low-absence students, with a median of 80 visits versus 22 (Appendix 6.2.2).

The 95% confidence interval for the mean difference was 29.20 to infinity (Appendix 6.2.3). Thus, students with low absenteeism averaged at least 29.20 more course content visits than those with high absenteeism.

Overall, the results provide strong evidence that students with fewer absences had higher course content engagement.

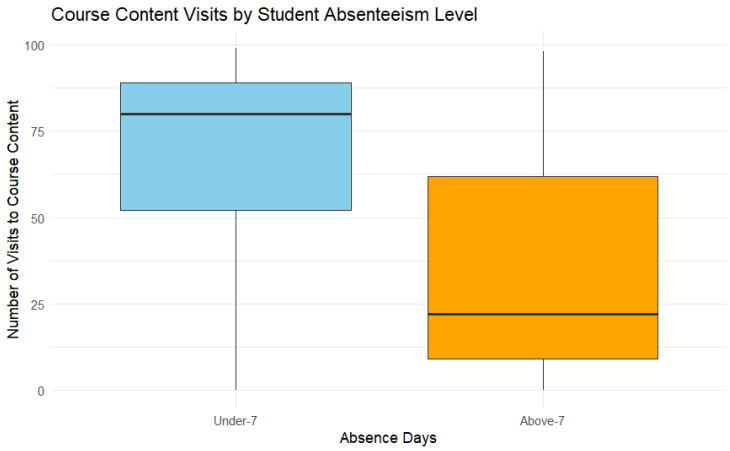


Figure 2. Box-Plots of Course Content Visits by Student Absenteeism Level

4.3 Guiding Question 3

The hypotheses are stated in Section 3.3(c). A bootstrap with 5,000 resamples within each absenteeism group and a Welch two-sample t-test were conducted at $\alpha = 0.05$.

Students with low absenteeism (<7 days, n = 289) averaged 48.19 discussion-post visits (SD = 27.13), compared with 35.85 visits (SD = 26.80) for students with high absenteeism (n = 191). The observed mean difference was 12.34 visits (Appendix 6.3.1).

The Welch t-test showed a significant difference, $t(410.3) = 4.91$, $p = 0.0000006$, so the null hypothesis was rejected. Figure 3 also shows higher engagement among low-absence students, with a median of 41 visits versus 30 (Appendix 6.3.2).

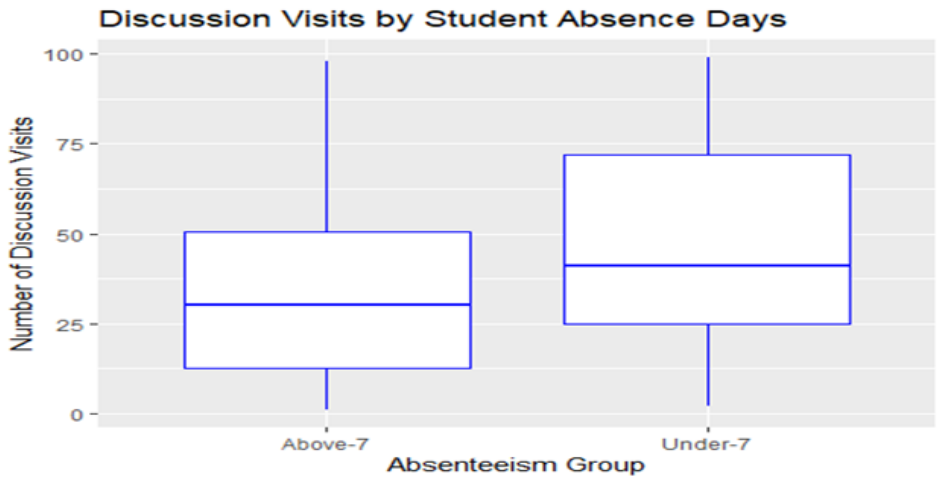


Figure 3. Box-Plots of Discussion Visits by Student Absence Days

The 95% bootstrap confidence interval for the mean difference was 7.39 to 17.25 visits (Appendix 6.3.4). Thus, students with low absenteeism averaged approximately 7.39–17.25 more discussion-post visits than those with high absenteeism.

Overall, the results provide strong evidence that students with fewer absences had higher discussion-post engagement.

4.4 Guiding Question 4

Let σ_{LH} = standard deviation of the number of hands raised among students with low absenteeism.
Let σ_{HH} = standard deviation of the number of hands raised among students with high absenteeism.

H_0: $\sigma_{LH} = \sigma_{HH}$

H_a: $\sigma_{LH} \neq \sigma_{HH}$

We will be using a significance level of $\alpha = 0.05$.

We will use Levene's Test of Equal Variance, with the median used as the location parameter. This test is essentially a two-independent-sample t-test applied to the absolute deviations of each observation from the median. This allows us to compare the variability between the two groups while being less sensitive to departures from normality than the traditional F-test. We require 2 assumptions to be satisfied in order for this test to be administered.

1. Independence: We are assuming independence as there is no verifiable way to prove if the data is dependent or independent, as there are no identifying columns.
2. Representative sample: The sample will be assumed to be representative of the target population of all Kalboard 360 students.

Performing the Levene's test of equal variance, we achieve a p-value of 0.09431 using the median as the center. Since 0.09431 > 0.05, we fail to reject the null hypothesis. There is insufficient evidence to conclude that the population standard deviations of raised hands differ between students with low and high absenteeism. (See appendix 6.4.1)

As shown in Figure 4, both groups have a similar overall spread. The Under-7 group has an IQR of approximately 50, while the Above-7 group has an IQR of approximately 40. Although the IQRs differ slightly, the box plot visually supports Levene's test conclusion that there is insufficient evidence of a difference in variability.

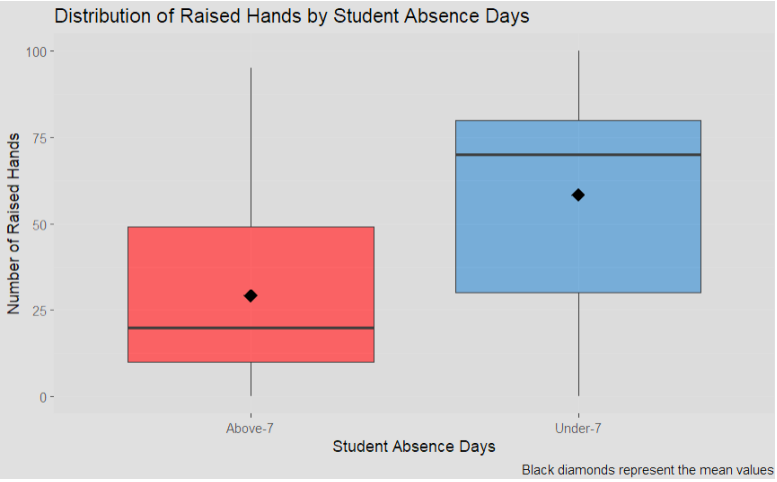


Figure 4. Boxplot of Above-7 and Under-7 against the number of raised hands (See appendix 6.4.2)

5. Conclusions

5.1 Summary

This study examined the relationship between student absenteeism, engagement, and academic performance using the xAPI-Edu-Data dataset of 480 students. The four research questions were analyzed using a permutation test, Welch two-sample t-tests, and Levene's test of equal variance, all at a significance level of $\alpha = 0.05$.

Guiding Question	**Sample statistics (Low vs High)**	**Test result**	**Conclusion**
High Performance	47.75% vs 2.09%	Permutation test; P = 0.0005; adjusted 95% CI [39.01%, 51.34%]	Strong evidence of an association between lower absenteeism and a higher proportion of high-level performers
Resource Use	68.20 vs 34.51	Welch t-test; p < 2.2e-16; 95% CI [29.20, Inf]	Strong evidence of an association between lower absenteeism and higher average course content visits.
Discussion Visits	48.19 vs 35.85	Welch t-test; p-value=0.0000006458 ;95% CI [8.200408, Inf]	Strong evidence that students with lower absenteeism tend to engage more with course discussion
Variability In Hands Raised	25.86 vs 28.21	Levene's Test for Homogeneity of Variance; F = 2.8103 p -value = 0.09431	Weak evidence that the standard deviation of the number of hands raised are different

5.2 Story

Taken together, the four analyses showed that lower absenteeism was associated with differences in both academic performance and measures of student engagement.

Most notably, low absenteeism was associated with a substantially higher proportion of high-level performers, 33.69 more course-content visits on average, and 12.34 more discussion visits on average.

At the same time, there was no statistically significant evidence of a difference in the variability of raised-hand participation. This pattern indicates that absenteeism was related to the level of performance and engagement, but not to the variability of participation.

5.3 Limitations and Future Work

A key limitation is that StudentAbsenceDays in xAPI-Edu-Data is provided only as two categories, Under-7 and Above-7. Consequently, the analyses cannot identify a dose-response relationship between the exact number of absences and academic performance or engagement. The dataset is also observational, so the observed associations cannot establish causation. In addition, other factors not included in the dataset may contribute to differences in student performance and engagement.

Future work could use datasets containing the exact number of absence days and additional student-level variables. Regression or other multivariable methods could then be used to account for potential confounding factors and investigate whether absenteeism remains associated with academic performance and engagement after adjustment for other relevant characteristics.

6. References

[1] E. A. Amrieh, T. Hamtini, and I. Aljarah, “Preprocessing and analyzing educational data set using X-API for improving student's performance,” in *2015 IEEE Jordan Conference on Applied Electrical Engineering and Computing Technologies (AEECT)*, Nov. 2015, pp. 1–5.

[2] E. A. Amrieh, T. Hamtini, and I. Aljarah, “Mining educational data to predict student’s academic performance using ensemble methods,” *International Journal of Database Theory and Application*, vol. 9, no. 8, pp. 119–136, 2016.

[3] Kanncaa1, “Factors affecting success in school,” *Kaggle*, [Online]. Available: https://www.kaggle.com/code/kanncaa1/factors-affecting-success-in-school

Appendix

6.1.1 Absence days vs class table

```
student_data = read.csv("xAPI-Edu-Data.csv")
dim(student_data)
table(student_data$StudentAbsenceDays, student_data$Class)
            H   L   M
  Above-7   4 116  71
  Under-7 138  11 140
```

6.1.2 Permutation test

```
# Creating a 1/0 data vector for High band performers
student_data = student_data %>% mutate(high_perf = ifelse(Class ==
"H", 1,0))
table(student_data$high_perf) # verifying the column and its values

 0   1
338 142

stats_gq1 = favstats(~high_perf | StudentAbsenceDays, data =
student_data)
stats_gq1
```

StudentAbsenceDays <chr>	min <dbl>	Q1 <dbl>	median <dbl>	Q3 <dbl>	max <dbl>	mean <dbl>	sd <dbl>	n <int>	missing <int>
Above-7	0	0	0	0	1	0.02094241	0.1435679	191	0
Under-7	0	0	0	1	1	0.47750865	0.5003603	289	0

```
obs_diff = stats_gq1$mean[2] - stats_qq1$mean[1]
print(obs_diff)

[1] 0.4565662
```

```
n_total = nrow(student_data)
n_under = 289
N = 1999
perm.outcome = numeric(N)

for(i in 1:N)
{
   index = sample(n_total, n_under, replace = FALSE)
   perm.outcome[i] = mean(student_data$high_perf[index]) -
mean(student_data$high_perf[-index])
}

perm.df = data.frame(perm.outcome)
summary(perm.df)

perm.outcome
 Min.   :-0.1608362
 1st Qu.:-0.0303991
 Median : 0.0043841
 Mean   :-0.0003313
 3rd Qu.: 0.0304716
 Max.   : 0.1348213
```

6.1.3 P-value and confidence interval

```
# Compute p-value
emp_pvalue_gq1 = (sum(perm.outcome >= obs_diff) + 1) / (N + 1)
emp_pvalue_gq1
[1] 5e-04

# Computing the 95% confidence-interval for the difference in the
proportions

x_HN = 138
x_HY = 4

n_HN = 289
n_HY = 191

prop.test(c(x_HN +1, x_HY + 1) , c(n_HN + 2, n_HY + 2), alternative =
"two.sided", correct = FALSE)$conf
[1] 0.3901454 0.5133676
attr(,"conf.level")
[1] 0.95
```

6.1.4 Histogram

```
ggplot(perm.df, aes(x = perm.outcome)) + geom_histogram(aes(y =
after_stat(density)), col = 'black', fill = 'steelblue', binwidth =
0.01) + geom_vline(xintercept = obs_diff, col = "red") +
xlab("Difference in Proportion of High Performers (Under-7 minus
Above-7)") +  ylab("Density") +
+  ggtitle("Permutation Distribution of the Difference in
Proportions")
```

6.2.1 Boxplot

```
ggplot(df, aes(x = factor(StudentAbsenceDays, levels = c("Under-7",
"Above-7")), y = VisITedResources, fill = StudentAbsenceDays)) +
geom_boxplot() + labs(title = "Course Content Visits by Student
Absenteeism Level", x = "Absence Days", y = "Number of Visits to
Course Content") + theme_minimal() + scale_fill_manual(values =
c("Under-7" = "skyblue", "Above-7" = "orange")) +
theme(legend.position = "none")
```

6.2.2 Welch's two-sample t-test

```
low_absenteeism = subset(df, StudentAbsenceDays == "Under-
7")$VisITedResources
high_absenteeism = subset(df, StudentAbsenceDays == "Above-
7")$VisITedResources

t_test_results = t.test(x = low_absenteeism, y = high_absenteeism,
alternative = "greater", var.equal = FALSE)
t_test_results

##Welch Two Sample t-test

##data:  low_absenteeism and high_absenteeism
##t = 12.356, df = 380.51, p-value < 2.2e-16
##alternative hypothesis: true difference in means is greater than 0
##95 percent confidence interval:
## 29.19516      Inf
##sample estimates:
##mean of x mean of y
## 68.20415  34.51309
```

6.3.1 Statistical Summary

```
 favstats(~Discussion | StudentAbsenceDays, data = student.df)
```

```
##   StudentAbsenceDays min   Q1 median   Q3 max      mean       sd   n
missing
 ## 1            Above-7   1 12.5     30 50.5  98 35.85340 26.79964 191
                                                                      0
 ## 2            Under-7   2 25.0     41 72.0  99 48.19377 27.12543 289
```

6.3.2 Side By Side Box Plot

```
library(ggplot2)
 ggplot(student.df, aes(x = StudentAbsenceDays, y = Discussion)) +
   geom_boxplot(col = "blue") +  xlab("Absenteeism Group") +
   ylab("Number of Discussion Visits") +
   ggtitle("Discussion Visits by Student Absence Days")
```

6.3.3 Welch's two-sample t-test

```
low_absence = student.df$Discussion[student.df$StudentAbsenceDays ==
"Under-7]
 high_absence = student.df$Discussion[student.df$StudentAbsenceDays ==
"Above-7"]t_test_result = t.test(low_absence,  high_absence,
alternative = "greater",
   conf.level = 0.95)

 t_test_result
```

```
##
 ##  Welch Two Sample t-test
 ##
 ## data:  low_absence and high_absence
 ## t = 4.9141, df = 410.3, p-value = 0.0000006458
 ## alternative hypothesis: true difference in means is greater than 0
 ## 95 percent confidence interval: ##  8.200408  Inf
 ## sample estimates:
 ## mean of x mean of y
 ##  48.19377  35.85340
```

6.3.4 Bootstrap Confidence Interval

```
boot_ci = qdata(  ~boot.diff,  c(0.025, 0.975),  data =
bootstrap.diff)
 boot_ci
```

```
##  2.5%     97.5%
 ##  7.394806 17.249144
```

6.4.1 Levene's Test

```
xAPI_Edu_Data = read.csv("xAPI-Edu-Data.csv")
High_absent_hands <-
xAPI_Edu_Data$raisedhands[xAPI_Edu_Data$StudentAbsenceDays == 'Above-
7']
low_absent_hands <-
xAPI_Edu_Data$raisedhands[xAPI_Edu_Data$StudentAbsenceDays == 'Under-
7']

head(high_absent_hands,4)
[1] 10 30 40 42
head(low_absent_hands,4)
[1] 15 20 50 12

favstats(high_absent_hands)
```

min <dbl>	Q1 <dbl>	median <dbl>	Q3 <dbl>	max <dbl>	mean <dbl>	sd <dbl>	n <int>	missing <int>
0	10	20	49	95	29.23037	25.85566	191	0

```
favstats(low_absent_hands)
```

min <dbl>	Q1 <dbl>	median <dbl>	Q3 <dbl>	max <dbl>	mean <dbl>	sd <dbl>	n <int>	missing <int>
0	30	70	80	100	58.37024	28.20556	289	0

```
leveneTest(raisedhands ~ StudentAbsenceDays, data = xAPI_Edu_Data,
conf.level = 0.95)
Levene's Test for Homogeneity of Variance (center = median: 0.95)
       Df F value  Pr(>F)
group   1  2.8103 0.09431 .
      478
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

6.4.2 Box and Whisker plot

```
ggplot(xAPI_Edu_Data,aes(x = StudentAbsenceDays, y = raisedhands, fill
= StudentAbsenceDays)) +
geom_boxplot(alpha = 0.7, outlier.colour = "red", outlier.shape = 16)
+
```

```
stat_summary(fun = mean, geom = "point", shape = 18, size = 4, color =
"black") +
scale_fill_manual(values = c("Above-7" = "red", "Under-7" = "blue")) +
labs(
    title = "Distribution of Raised Hands by Student Absence Days",
    x = "Student Absence Days",
    y = "Number of Raised Hands",
    caption = "Black diamonds represent the mean values"
  ) +
  theme(legend.position = "none")
```